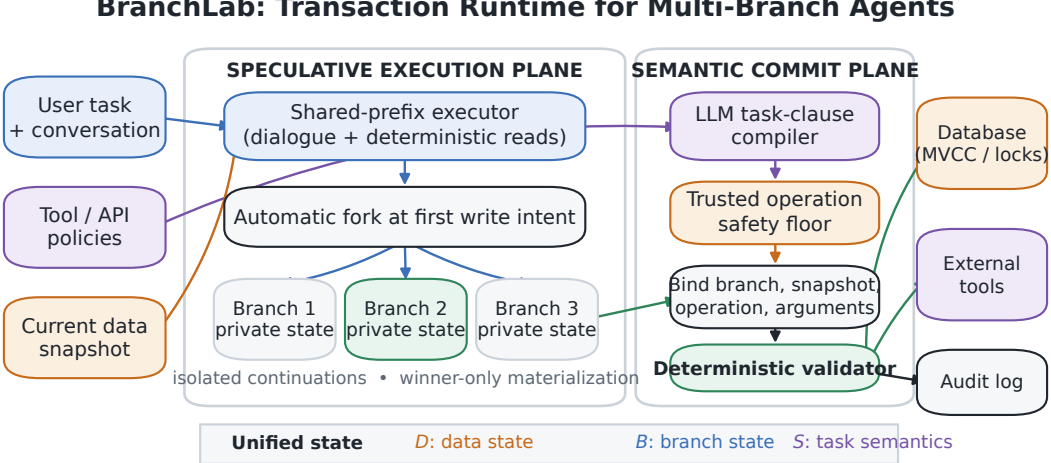

BranchLab: Transaction Runtime for Multi-Branch Agents
SPECULATIVE EXECUTION PLANE
SEMANTIC COMMIT PLANE
User task + conversation
Tool / API policies
Current data snapshot
Shared-prefix executor (dialogue + deterministic reads)
Automatic fork at first write intent
Branch 1 private state
Branch 2 private state
Branch 3 private state
isolated continuations • winner-only materialization
LLM task-clause compiler
Trusted operation safety floor
Bind branch, snapshot, operation, arguments
Deterministic validator
Database (MVCC / locks)
External tools
Audit log
Unified state
D: data state
B: branch state
S: task semantics